

# Jeehong Kim

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## RESEARCH INTEREST

I am interested in designing broadly applicable machine learning techniques for graph-based problems, especially in heterophilous and anomalous scenarios.

- **Keywords:** Graph Representation Learning, Graph Heterophily, Anomaly Detection, GNNs with LLM

## EDUCATION

**Seoul National University**, Seoul, South Korea

- M.S.-Ph.D. Integrated Program in Data Science Mar 2023 – Present
  - Research Interest: Graph Representation Learning, Graph Heterophily, Anomaly Detection
  - Advisor: Prof. Hyunwoo Park
- Planning to fulfill military service as a *Technical Research Personnel* starting around September 2026

**Korea University**, Seoul, South Korea

- B.S. in Industrial Management Engineering Mar 2019 – Feb 2023
  - GPA: 3.76 / 4.50

## RESEARCH EXPERIENCE

**Materials Intelligence Lab**, LG AI Research

- Research Intern Feb 2026 – Present
  - Research Interest: Molecule Property Prediction, Graph Neural Network

**DSAIL (Data Science & Artificial Intelligence)**, KAIST

- Undergraduate Research Intern Jun 2022 – Aug 2022
  - Research Interest: Graph Neural Network, Recommendation System
  - Advisor: Prof. Chanyoung Park

**AIDA Lab (Artificial Intelligence & Data Analytics)**, Korea University

- Undergraduate Research Intern May 2021 – Jan 2022
  - Research Interest: Anomaly Detection, GAN
  - Advisor: Prof. Sungwon Han

## PUBLICATIONS

(†: Equal contribution)

### JOURNALS

- [J2] STGVAD: Spatio-Temporal Graph-based Vessel Behavior Anomaly Detection  
**Jeehong Kim**<sup>†</sup>, Minchan Kim<sup>†</sup>, Youngseok Hwang, Sungho Bae, Deuk Jae Cho, Wonhee Lee, Hyunwoo Park  
**IEEE Access** (2025.09)
- [J1] FURIOUS: Fully unified risk-assessment with interactive operational user system for vessels  
Yooyeun Kim<sup>†</sup>, **Jeehong Kim**<sup>†</sup>, Wonhee Lee, Hyunwoo Park, Deuk Jae Cho  
**PLoS One** (2025.05)

### WORKSHOPS

- [W2] Spatio-Temporal Graphs Beyond Grids: Benchmark for Maritime Anomaly Detection  
**Jeehong Kim**<sup>†</sup>, Youngseok Hwang<sup>†</sup>, Minchan Kim, Sungho Bae, Hyunwoo Park  
**NeurIPS 2025 Workshop in AI for Science: The Reach and Limits of AI for Scientific Discovery**
- [W1] Adaptive Sparsified Graph Learning Framework for Vessel Behavior Anomalies  
**Jeehong Kim**<sup>†</sup>, Minchan Kim<sup>†</sup>, Jaeseong Ju, Youngseok Hwang, Wonhee Lee, Hyunwoo Park  
**AAAI 2025 Workshop in Anomaly Detection in Scientific Domains**

## PROJECTS

### MOIN

- Fintech Technology Commercialization Support Program Jul 2025 – Present
  - Fraud detection system (FDS) modeling optimization for overseas remittance using AI technology

**Korea Research Institute of Ships & Ocean Engineering (KRISO)**

- Vessel Behavior Anomaly Detection Oct 2024 – Present
  - Anomaly detection using a GNN-based framework on a customized dataset
  - (Published) *Adaptive Sparsified Graph Learning Framework for Vessel Behavior Anomalies*
  - (Published) *STGVAD: Spatio-Temporal Graph-based Vessel Behavior Anomaly Detection*
- Visualization for Ship Collision Risk Assessment Mar 2024 – May 2025
  - Development of a decision supportive interactive system for ship collision avoidance
  - (Published) *FURIOUS: Fully unified risk-assessment with interactive operational user system for vessels*

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                    |                                            |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
|                                  | <b>VAIV Company</b> <ul style="list-style-type: none"> <li>Generative AI Professional Development Project <ul style="list-style-type: none"> <li>Task planning with LLM-guided GNNs</li> </ul> </li> </ul>                                                                                                                                                                                                         | Mar 2024 – Present                         |
|                                  | <b>Seoul National University</b> <ul style="list-style-type: none"> <li>SNU Creative Education Project (Inno-Edu 2031) <ul style="list-style-type: none"> <li>Curriculum reform initiative under the SNU Creative Education Project</li> </ul> </li> <li>Undergraduate Course Development <ul style="list-style-type: none"> <li>(L0655.005100) Veritas Practice: Data Visualization and Me</li> </ul> </li> </ul> | Jan 2024 – Dec 2024<br>Jan 2024 – Jun 2024 |
| <b>AWARDS &amp; SCHOLARSHIPS</b> | <b>Best Poster Award</b> <ul style="list-style-type: none"> <li>NeurIPS 2025 Workshop in AI for Science: The Reach and Limits of AI for Scientific Discovery</li> </ul>                                                                                                                                                                                                                                            | Dec 2025                                   |
|                                  | <b>BK 21 Scholarship</b> , Brain Korea                                                                                                                                                                                                                                                                                                                                                                             | Mar 2025 – Present                         |
|                                  | <b>SNU Medical AI Scholarship</b> , Seoul National University                                                                                                                                                                                                                                                                                                                                                      | Aug 2024 – Present                         |
|                                  | <b>Veritas Scholarship</b> , Korea University <ul style="list-style-type: none"> <li>Research on detection of anomalies in the manufacturing process <ul style="list-style-type: none"> <li>Advisor: Prof. Sungwon Han</li> </ul> </li> </ul>                                                                                                                                                                      | Feb 2022                                   |
|                                  | <b>HAICON2021</b> , Korea Institute of Information Security & Cryptology (KIISC) <ul style="list-style-type: none"> <li>Construction of an AI model for identifying time-series based anomalies in industrial control systems <ul style="list-style-type: none"> <li>Ranked 6th out of 177 teams</li> </ul> </li> </ul>                                                                                            | Nov 2021                                   |
|                                  | <b>Industrial Engineering Project Competition</b> , Korean Institute of Industrial Engineers (KIIE) <ul style="list-style-type: none"> <li>Design of a dynamic corridor network for K-UAM <ul style="list-style-type: none"> <li>Received participation award in industrial engineering project competition</li> </ul> </li> </ul>                                                                                 | Oct 2021                                   |
|                                  | <b>Data Visualization Competition</b> , Korea University, Yonsei University, Intel <ul style="list-style-type: none"> <li>Visualization of patient and their deductible premium relationship <ul style="list-style-type: none"> <li>Received grand prize</li> </ul> </li> </ul>                                                                                                                                    | Aug 2021                                   |
| <b>PROFESSIONAL SERVICES</b>     | <b>Conference Review</b> <ul style="list-style-type: none"> <li>Learning on Graphs Conference (LOG)</li> </ul>                                                                                                                                                                                                                                                                                                     | 2025                                       |
|                                  | <b>Workshop Review</b> <ul style="list-style-type: none"> <li>The Reach and Limits of AI for Scientific Discovery (AI4Science)@NeurIPS 2025</li> </ul>                                                                                                                                                                                                                                                             | 2025                                       |
|                                  | <b>Journal Review</b> <ul style="list-style-type: none"> <li>IEEE Access</li> </ul>                                                                                                                                                                                                                                                                                                                                | 2025                                       |
| <b>ACTIVITIES</b>                | <b>Lab Captain</b> , Visualization and Business Analytics Lab (ViBA) <ul style="list-style-type: none"> <li>Advisor: Prof. Hyunwoo Park</li> </ul>                                                                                                                                                                                                                                                                 | Dec 2024 – Present                         |
|                                  | <b>Samsung Dream Class</b> , Samsung Welfare Foundation                                                                                                                                                                                                                                                                                                                                                            | Sep 2021 – Feb 2022                        |
|                                  | <b>Korea-China Leadership Program</b> , Korea Foundation for Advanced Studies (KFAS)                                                                                                                                                                                                                                                                                                                               | Jul 2019                                   |
| <b>TEACHING EXPERIENCE</b>       | <b>Teaching Assistant</b> <ul style="list-style-type: none"> <li>K-Digital training program at Seoul National University</li> <li>Python data analysis training program for Seoul National University faculty members</li> </ul>                                                                                                                                                                                   | Jun 2024<br>Jan 2024                       |
| <b>TALKS AND SEMINARS</b>        | <b>FURIOUS: Fully unified risk-assessment with interactive operational user system for vessels</b> <ul style="list-style-type: none"> <li>2024 INFORMS Annual Meeting (Contributed Session)</li> </ul>                                                                                                                                                                                                             | Oct 2024                                   |
|                                  | <b>Spatio-Temporal Dynamics-Aware GNN Framework for Vessel Behavior Anomaly Detection</b> <ul style="list-style-type: none"> <li>2025 Korea Data Mining Society (Best Paper Award)</li> </ul>                                                                                                                                                                                                                      | Aug 2025                                   |
| <b>REFERENCES</b>                | <b>Prof. Hyunwoo Park</b> , Associate Professor, Seoul National University <ul style="list-style-type: none"> <li>Email: hyunwoopark@snu.ac.kr</li> </ul>                                                                                                                                                                                                                                                          |                                            |

[CV compiled on 2026-01-20]